Introduction

Jensen's inequality relates the expectation of a function to the function of an expectation: for a convex function g, $E[g(X)] \geq g(E[X])$, with equality only when X is a constant (or g is linear). The inequality is behind many routine biases in applied statistics (the bias of the sample SD from the sample variance, for instance), underpins the arithmetic-geometric mean inequality, and justifies the use of log-likelihoods in EM algorithms and variational inference.

Prerequisites

The reader should know what a convex function is (or equivalently, what a concave function is) and the definition of an expectation.

Theory

A function $g : \mathbb{R} \to \mathbb{R}$ is **convex** if for any a, b and $\lambda \in [0, 1]$,

$$g(\lambda a + (1-\lambda)b) \leq \lambda g(a) + (1-\lambda)g(b).$$

Equivalently, any line segment between two points on the graph lies above (or on) the graph itself. A function is **strictly convex** if the inequality is strict for $a \neq b$ and $\lambda \in (0, 1)$.

Jensen's inequality. If g is convex and X is integrable,

$$g(E[X]) \leq E[g(X)].$$

If g is strictly convex and X is non-degenerate, the inequality is strict. For concave g, the inequality flips: $g(E[X]) \geq E[g(X)]$.

Applications.

- **Log is concave**: $E[\log X] \leq \log E[X]$. Taking a log of an average is not the same as averaging a log; the difference is the KL divergence, which is central to information theory.
- **Square is convex**: $E[X^2] \geq (E[X])^2$; the variance is non-negative.
- **Reciprocal**: $E[1/X] \geq 1/E[X]$ for positive X. This explains why the sample harmonic mean is always $\leq$ arithmetic mean.
- **Exponential**: $E[e^X] \geq e^{E[X]}$. The MGF at $t = 1$ exceeds $e^{E[X]}$.

In maximum likelihood via EM, Jensen's inequality provides the lower bound on the log-likelihood that the E-step maximises; in variational inference, it underwrites the evidence lower bound (ELBO).

Assumptions

- X must be integrable: $E[|X|] < \infty$.
- g is convex (or concave). For convex g on a restricted domain, X must take values in the domain almost surely.

R Implementation

```r
set.seed(2026)
n <- 10000
x <- rlnorm(n, meanlog = 0, sdlog = 1)

mean_x <- mean(x)
log_of_mean <- log(mean_x)
```

```r
mean_of_log <- mean(log(x))

c(log_of_mean = log_of_mean,
  mean_of_log = mean_of_log,
  gap         = log_of_mean - mean_of_log)

theoretical_gap <- log(exp(0.5)) - 0   # = 0.5
theoretical_gap
```

For lognormal data, the gap $\log E[X] - E[\log X]$ is exactly $\sigma^2/2$ where σ is the lognormal scale. The simulation confirms.

Output & Results

```
log_of_mean  mean_of_log         gap
     0.499       -0.002       0.501

[1] 0.5
```

The difference is approximately 0.5 – exactly what Jensen predicts for this distribution. The gap is never negative, as the inequality requires.

Interpretation

Jensen's inequality explains why many estimators are biased after a transformation. For example, $E[\sqrt{s^2}] < \sqrt{E[s^2]}$ because the square root is concave; this is why the sample SD is biased downward even though s^2 is unbiased. Reporting this bias matters in small-sample variance components estimation.

Practical Tips

- Always check convexity before applying Jensen: the sign of the inequality flips for concave g.
- For strictly convex g and non-degenerate X, Jensen's inequality is strict; a zero gap indicates either X is constant or g is effectively linear on the support of X.
- Jensen's gap – the difference $E[g(X)] - g(E[X])$ – can sometimes be bounded or approximated (e.g., $\frac{g''(\mu)\sigma^2}{2}$ to first non-trivial order).
- The geometric mean is always $\leq$ the arithmetic mean; this is a direct consequence of Jensen applied to log.
- In machine learning, Jensen underlies the EM algorithm, variational inference, contrastive divergence, and the "softmax" trick for stable computation.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Scientific process and research workflow
- R, RStudio, Quarto, and renv toolchain
- Data types, tidy data, accuracy and precision

- Import, joins, and missingness with dplyr